

# Amazon S3 Assignment 1

## HOW TO CREATE AND CONFIGURE AN AMAZON S3 BUCKET

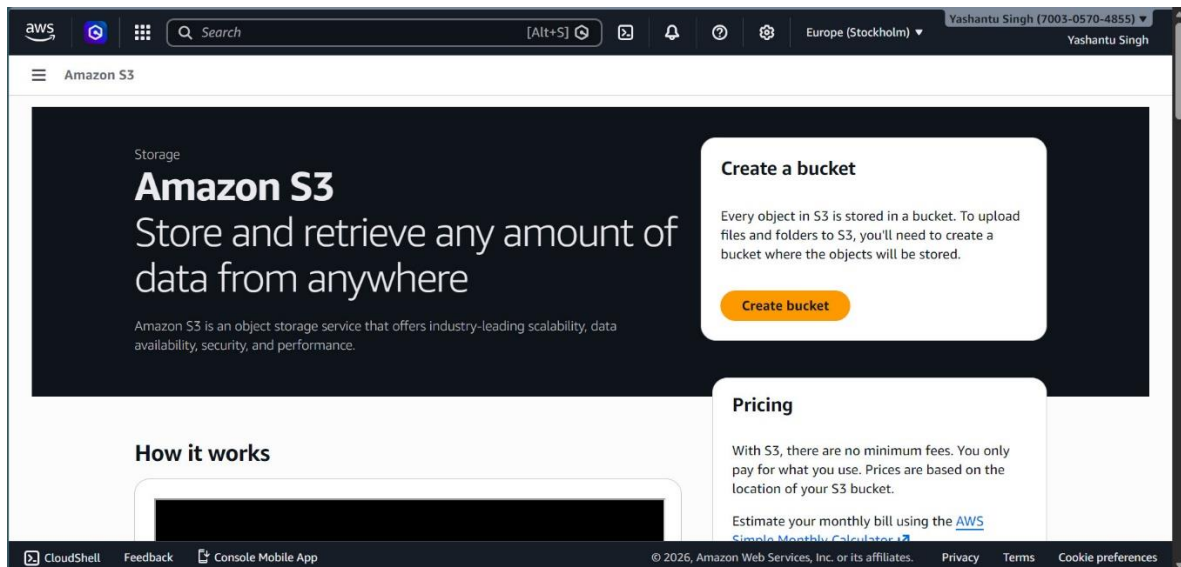
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### Introduction to Amazon S3 and buckets

Amazon Simple Storage Service (Amazon S3) is a scalable object storage service that allows users to store and retrieve any amount of data from anywhere. An S3 bucket is a container for storing objects (files), and each bucket is uniquely named within AWS. Buckets are used to organize and manage data in the cloud.

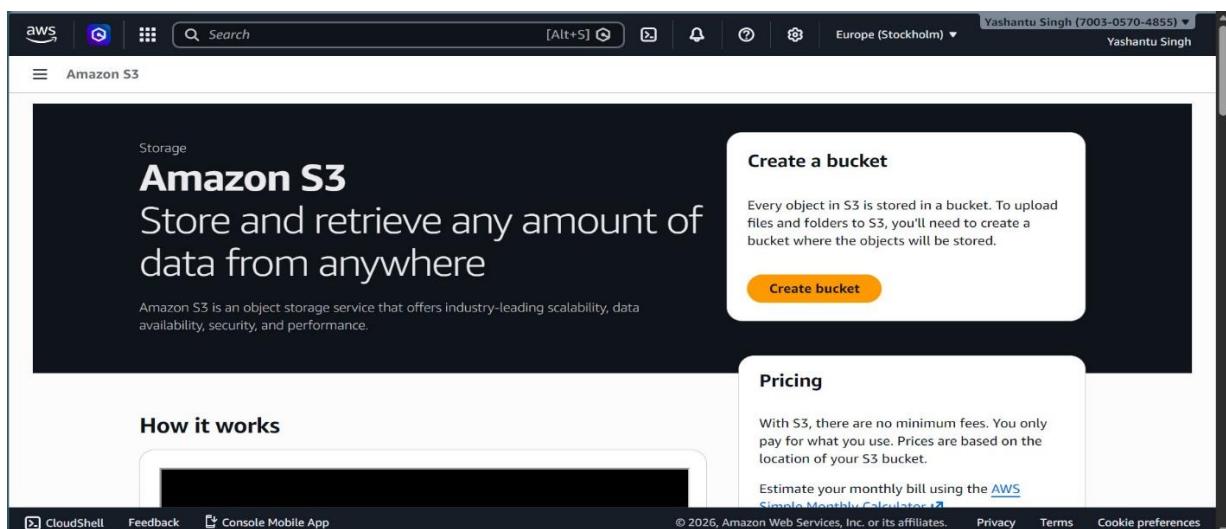
### Step 1: Navigate to Amazon S3 service in AWS Console

Log in to your AWS Management Console. From the AWS services menu, search for and select 'S3' to open the Amazon S3 dashboard.



### Step 2: Click 'Create bucket'

On the Amazon S3 dashboard, click the orange 'Create bucket' button to start the process of creating a new storage bucket.



## Step 3: Configure general settings (Region, Bucket type, Namespace, Bucket name)

Choose the AWS Region where your bucket will reside. Select the bucket type (General purpose or Directory). Choose the namespace (Global or Account Regional). Enter a unique bucket name following AWS naming conventions.

The screenshot shows the AWS console 'Create bucket' page. The 'General configuration' section is active. The 'AWS Region' is set to 'Europe (Stockholm) eu-north-1'. The 'Bucket type' is set to 'General purpose'. The 'Bucket namespace' is set to 'Global namespace'. The 'Bucket name' is 'amzn-s3-yashantu-singh'. The 'Object Ownership' section is also visible, with 'ACLs disabled (recommended)' selected.

**Create bucket** [Info](#)

Buckets are containers for data stored in S3.

**General configuration**

**AWS Region**  
Europe (Stockholm) eu-north-1

**Bucket type** [Info](#)

☒ **General purpose**  
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**  
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

**Bucket namespace**  
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**  
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**  
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

**Bucket name** [Info](#)

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

**Copy settings from existing bucket - optional**  
Only the bucket settings in the following configuration are copied.

Format: s3://bucket/prefix

**Object Ownership** [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

**Object Ownership**

☒ **ACLs disabled (recommended)**  
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**  
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

## Step 4: Set object ownership (ACLs disabled recommended)

Choose the object ownership setting. It is recommended to disable ACLs to ensure all objects are owned by the bucket owner and access is managed through policies.

This screenshot is a continuation of the previous one, showing the 'Object Ownership' section. The 'ACLs disabled (recommended)' option is selected, and the 'Choose bucket' button is visible.

**Bucket name** [Info](#)

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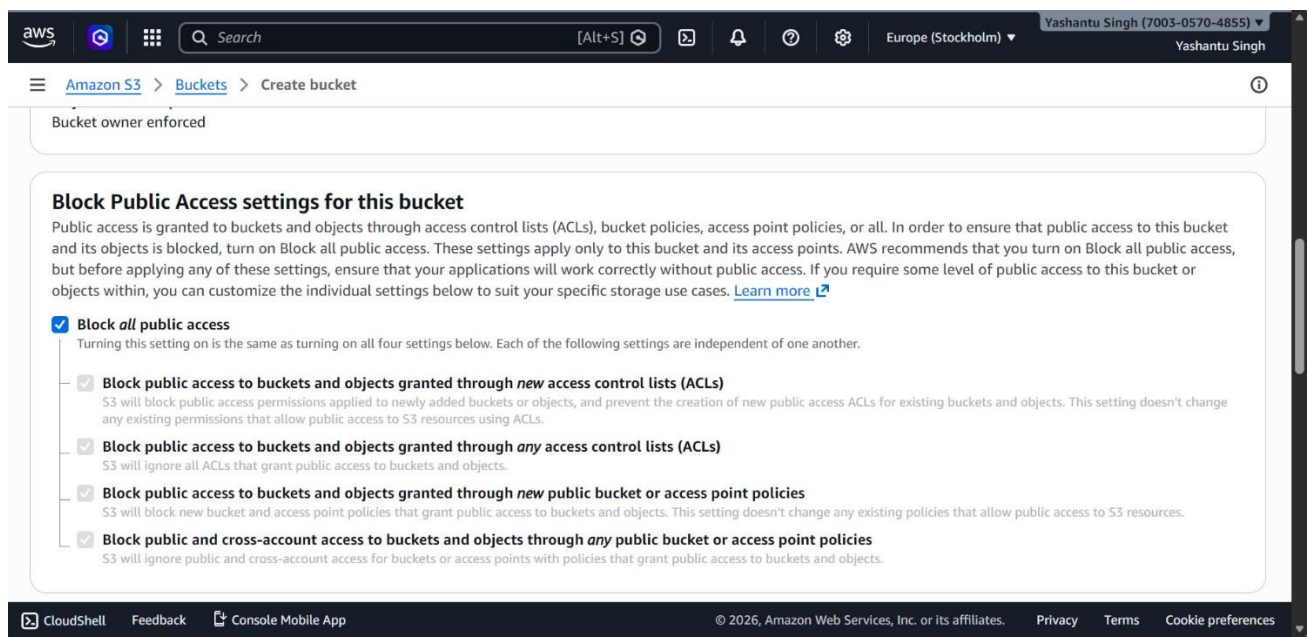
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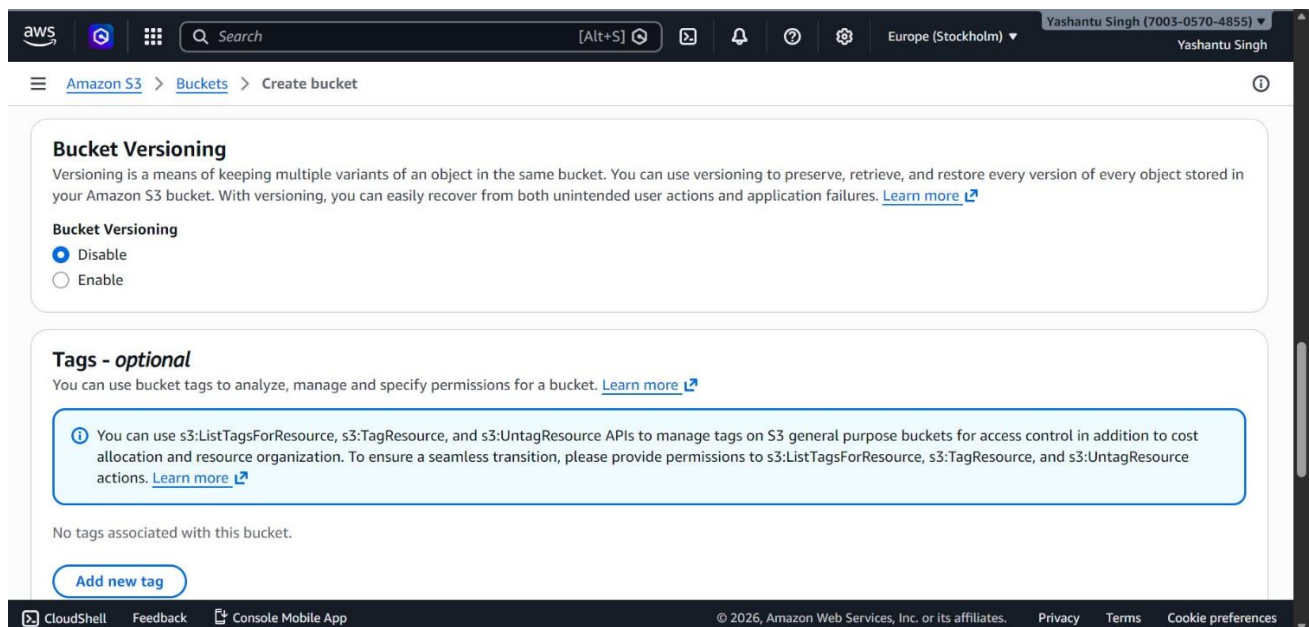
## Step 5: Configure public access settings (Block all public access recommended)

Enable 'Block all public access' to prevent unauthorized access to your data. This is a recommended best practice for securing your data.



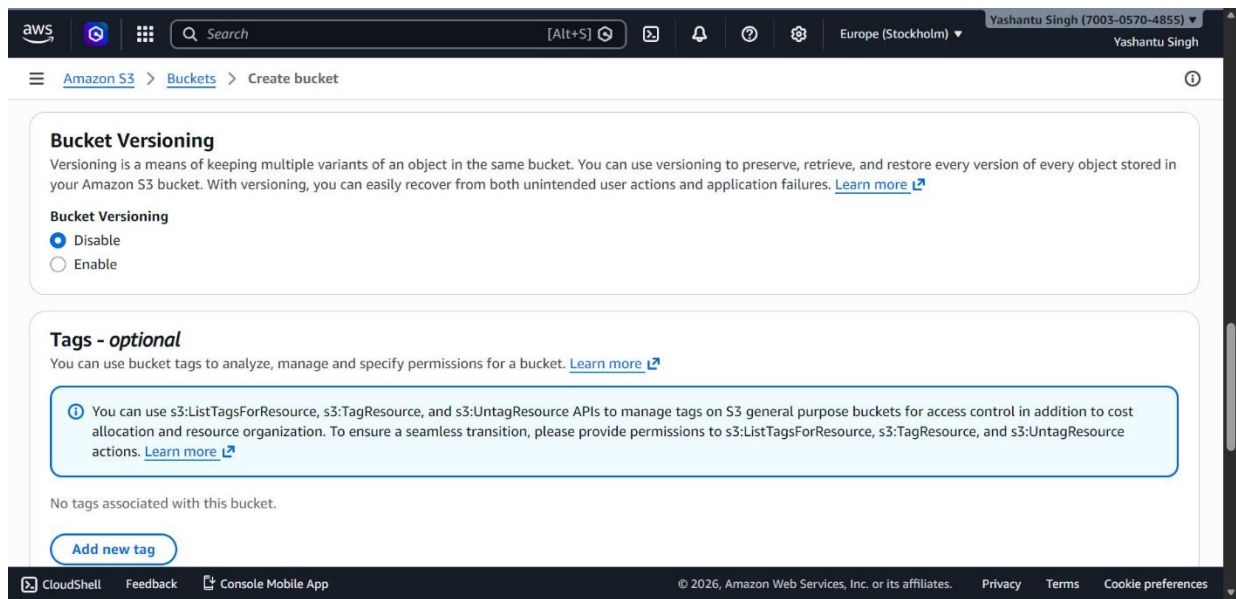
## Step 6: Enable or disable bucket versioning

Bucket versioning allows you to preserve, retrieve, and restore every version of every object store in your bucket. It is recommended to enable versioning to protect against accidental deletions or overwrites.



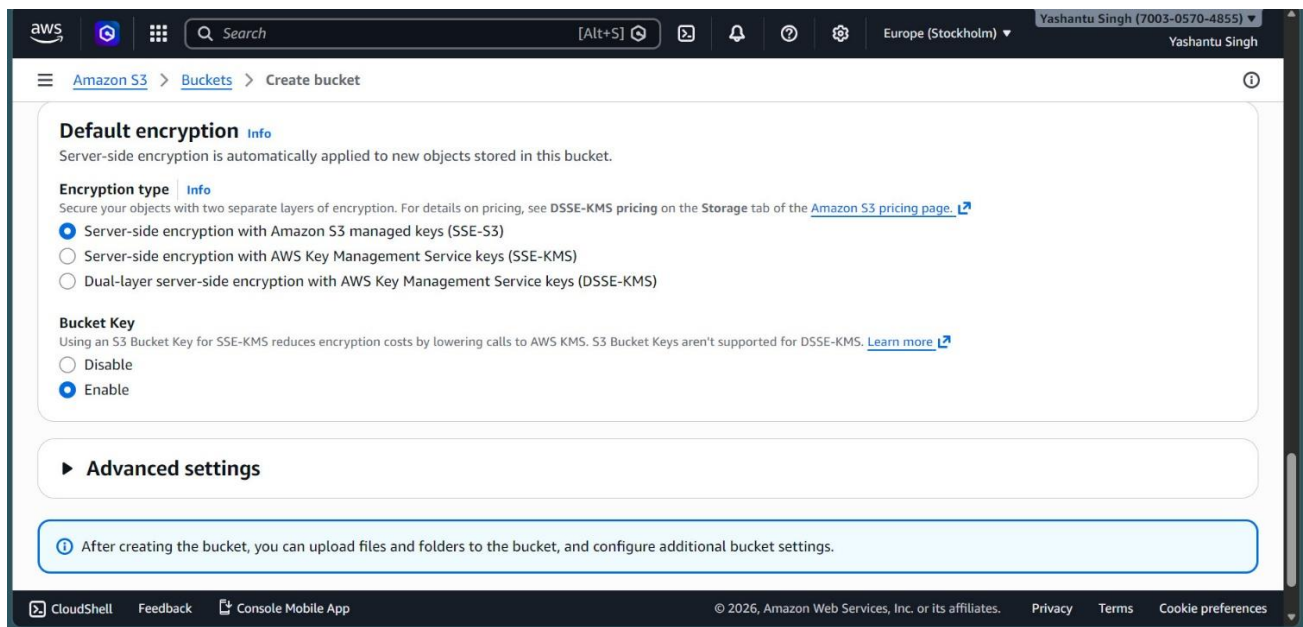
## Step 7: Add tags for organization and cost allocation

Tags are key-value pairs that help you organize and manage your AWS resources. They can also be used for cost allocation and access control.



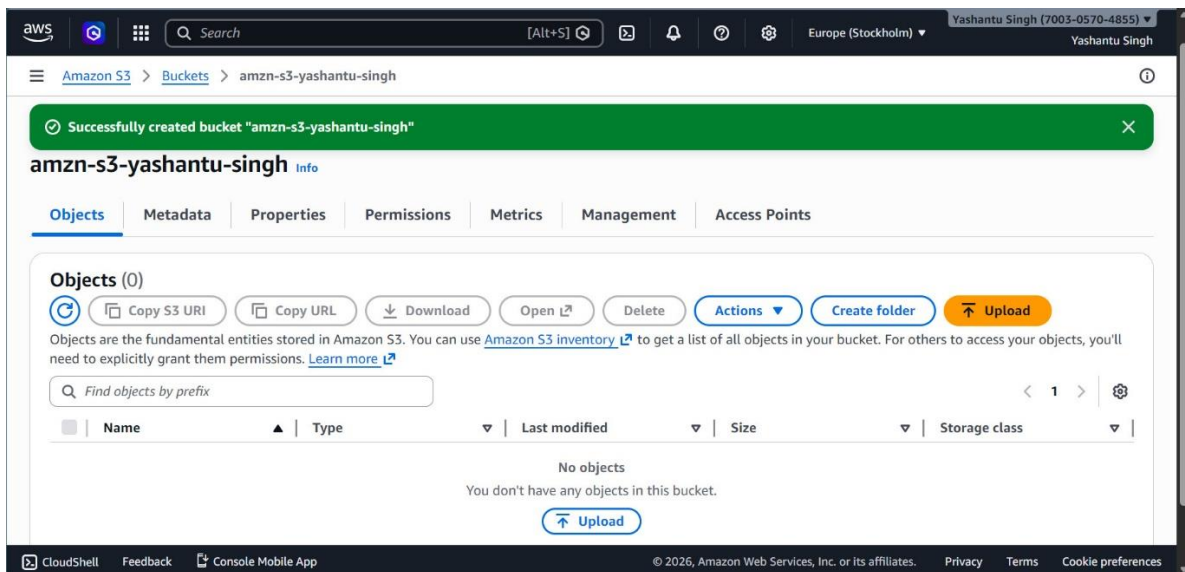
## Step 8: Configure default encryption (SSE-S3, SSE-KMS, or DSSE-KMS)

Choose a server-side encryption method to protect your data at rest. Options include SSE-S3 (default), SSE-KMS (with AWS Key Management Service), or DSSE-KMS (dual-layer encryption).



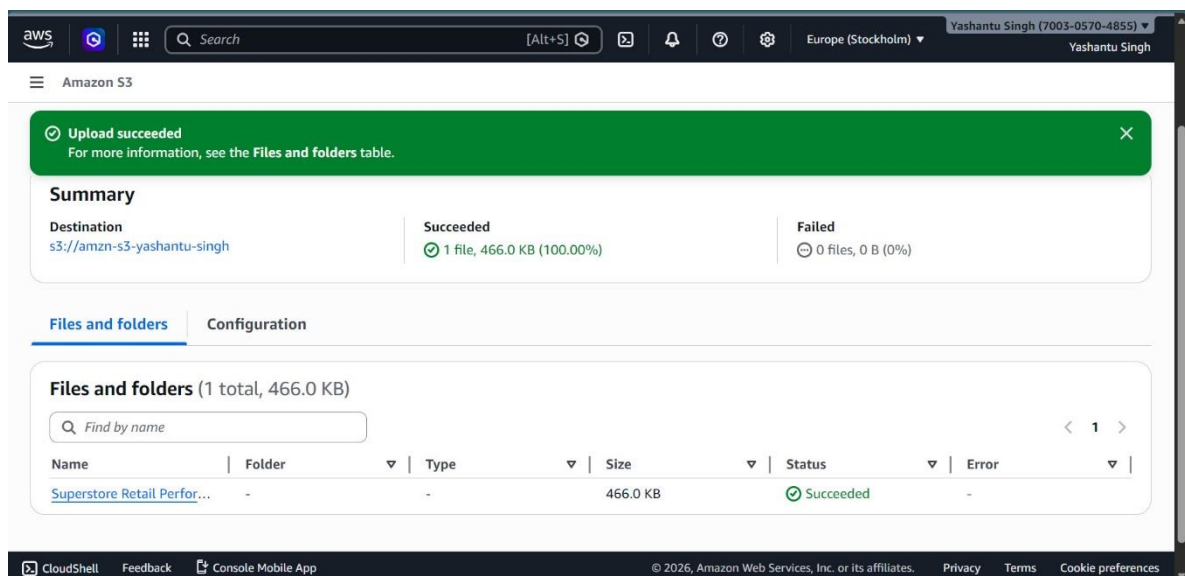
## Step 9: Review and create the bucket

Review all your configurations. If everything looks correct, click the 'Create bucket' button to finalize the creation.



## Step 10: Upload objects to the bucket and verify successful upload

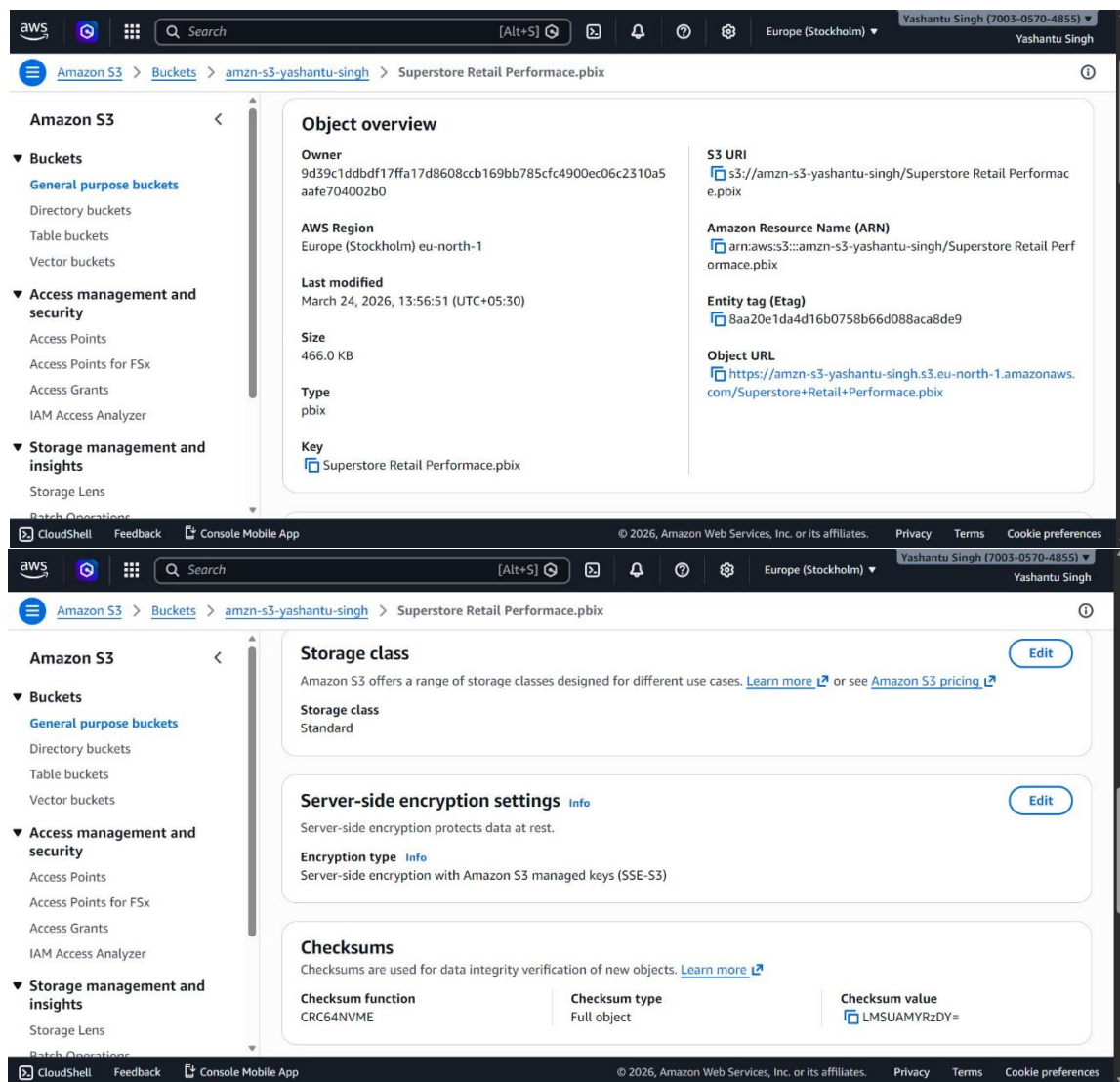
After creating the bucket, you can upload files using the 'Upload' button. Once uploaded, verify the success message and check the file list.





## Step 11: View object properties (metadata, ARN, S3 URI, checksum, encryption)

Click on the uploaded object to view its properties, including metadata, Amazon Resource Name (ARN), S3 URI, encryption settings, and checksum for data integrity.



## Step 12: Manage bucket lifecycle (versioning, replication, expiration rules)

Use lifecycle configurations to manage your objects. You can enable replication, set expiration rules, and manage versioning to optimize storage and data retention.

